Statistical analysis for musical corpora – Part 2

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In previous lecture

- We learned how to conceptualize a research question and operationalize it
- How to look at a single variable
  - Data distribution
  - Statistical moments
- Student's T-test

Case study

What is a hook in music?

A hook is a musical idea, often a short riff, passage, or phrase, that is used in popular music to make a song appealing and to "catch the ear of the listener”

Research questions

- Do all hooks have something in common (is there a common recipe to write a song that contains one)?
- Are the hooks usually found in some particular part of a song? The very beginning, may be? In a culmination? A chorus?
- Is there a required length for something to become a hook? Can it be an instrumental part or does it require lyrics?

How can we answer the questions?

Qualitative methods

- Ask prominent pop music composers
- Listen to some popular songs and describe the parts that you think sound most like a hook
- Ask music listeners which part of their favourite song they consider a hook and why

Quantitative methods

- Collect data from the listeners on which parts of a song are the hooks
- Measure the lengths of the hooks vs normal musical phrases
- Measure their loudness, repetition ratio, position in a song

Qualitative result

Hooks are 'the foundation of commercial songwriting, particularly hit-single writing'. Hooks may involve repetition of 'one note or a series of notes or of a lyric phrase, full lines or an entire verse'. The hook is 'what you're selling'. Though a hook can be something as insubstantial as a 'sound' (such as da doo ron ron), 'ideally it should contain one or more of the following:
(a) a driving, danceable rhythm;
(b) a melody that stays in people's minds;
(c) a lyric that furthers the dramatic action, or defines a person or place'

(Kasha and Hirschhorn 1979, pp. 28, 29)

Hooked: a Game with a Purpose

Hooked: game with a purpose

(a) Recognition (b) Verification (c) Prediction

Figure 1: Screenshots from the Hooked prototype. (a) The recognition task is the heart of the game: A song starts from the beginning of an internal musical section, chosen at random, and the player must guess the song as quickly as possible. (b) The sound then mutes for a few seconds while players try to follow along in their heads. When the sound comes back, players must verify that the song is playing back from the correct place. (c) Occasionally, players instead must do the reverse: predict which of two sections is catchiest, in order to store bonus rounds for themselves.

Burgoyne, J. A., Bountouridis, D., van Balen, J. M. H., & Honing, H. (2013). "Hooked: A Game for Discovering What Makes Music Catchy." Proceedings of the **International Society for Music Information Retrieval (ISMIR 2013)**.

Aljanaki, A., Bountouridis, D., Burgoyne, J. A., van Balen, J., Wiering, F., Honing, H., & Veltkamp, R. (2013/2014). "Designing Games with a Purpose for Data Collection in Music Research: Emotify and Hooked." Proceedings of the **Games and Learning Alliance Conference (GaLA)**.

What kind of data is generated?

- Binary data – a user knew or did not know the song
- Reaction time data – how fast did a user click yes
- Did they recognize the correct spot once the music came back?

Recognition time for Adele's 'Rumour Has It'

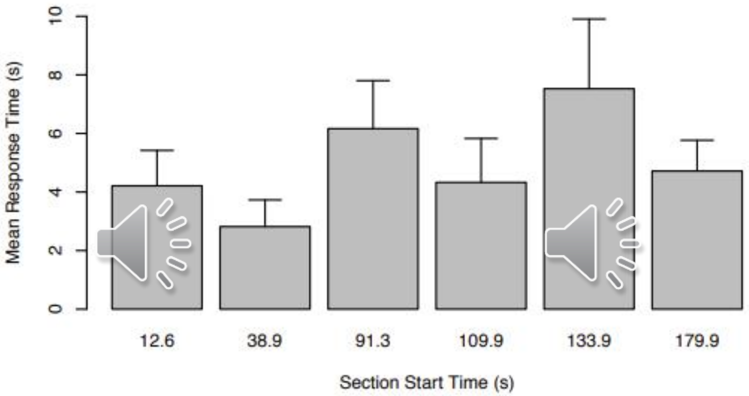


Figure 2: Mean response times from the recognition task on different sections of Adele's 'Rumour Has It'. Error bars reflect standard error. Controlling for multiple comparisons, there are significant differences ($p < .05$) in response times between the bridge (133.9 s) or the verse at 91.3 s, and the initial entry of the vocals (12.6 s) or the pre-chorus at 38.9 s.

Computational analysis of hooks

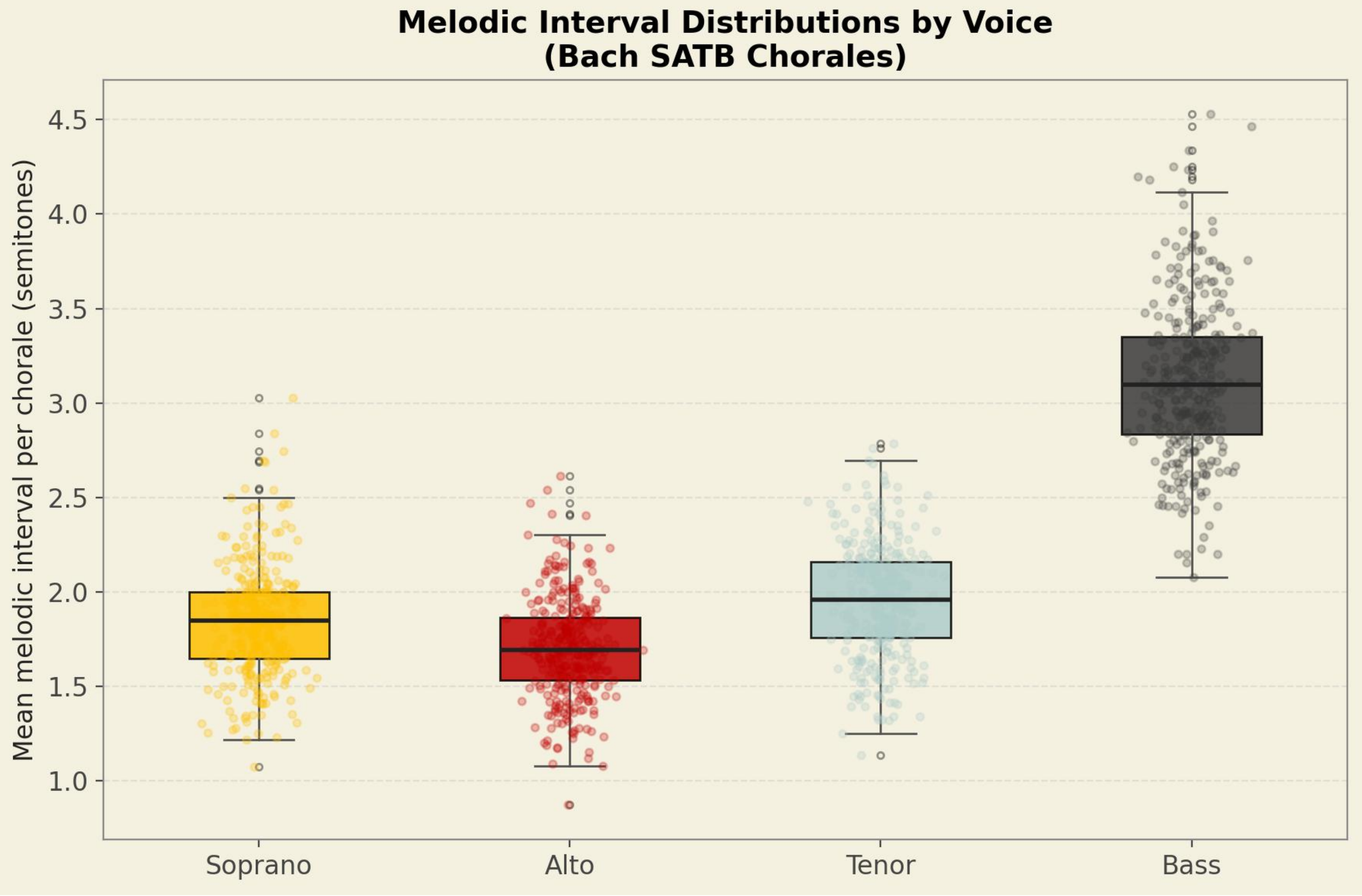
Parameter	Audio[a] $\hat{\beta}$	Audio[a] 99.5 % CI	Audio[b] $\hat{\beta}$	Audio[b] 99.5 % CI	Symbolic[b] $\hat{\beta}$	Symbolic[b] 99.5 % CI	Combined[b] $\hat{\beta}$	Combined[b] 99.5 % CI
Fixed effects								
Intercept	−0.84	[−0.91, −0.77]	−0.67	[−0.78, −0.56]	−0.62	[−0.73, −0.51]	−0.63	[−0.74, −0.53]
Audio								
Vocal Prominence	0.14	[0.10, 0.18]	0.11	[0.04, 0.17]			0.08	[0.01, 0.15]
Timbral Conventionality	0.09	[0.05, 0.13]						
Melodic Conventionality	0.06	[0.02, 0.11]						
M/H Entropy Conventionality	0.06	[0.02, 0.10]						
Sharpness Conventionality	0.05	[0.02, 0.09]						
Harmonic Conventionality	0.05	[0.01, 0.10]						
Timbral Recurrence	0.05	[0.02, 0.08]						
Mel. Range Conventionality	0.05	[0.01, 0.08]	0.07	[0.02, 0.13]			0.07	[0.01, 0.12]
Symbolic								
Melodic Repetitivity					0.12	[0.06, 0.19]	0.11	[0.05, 0.17]
Mel./Bass Conventionality					0.07	[0.01, 0.13]	0.08	[0.01, 0.14]

Van Balen, J; Burgoyne, J.A; Bountouridis, D; Müllensiefen, Daniel; and Veltkamp, R. 2015. 'Corpus Analysis Tools for Computational Hook Discovery.'. In: *Proceedings of the 16th International Society for Music Information Retrieval (ISMIR) Conference 2014*. Spain.

Statistics for music: Part 2

T-test: recap

- With t-test, we compared two groups – basses and sopranos
- We learned that basses had bigger melodic intervals


Melodic Interval Distributions by Voice
(Bach SATB Chorales)
Mean melodic interval per chorale (semitones)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Soprano
Alto
Tenor
Bass

Related samples t-test

- We are comparing two related samples (soprano and bass parts come from the same chorale)

```
import scipy
scipy.stats.ttest_rel(soprano_interval_means, bass_interval_means)
```

Reporting

A paired-samples t-test showed that soprano intervals (M = 1.85, SD = 0.3) were significantly smaller than bass intervals (M = 3.11, SD = 0.42), $t(319) = -46.05$, $p < .001$.

But we have more than 2 voices

Do we now do the paired t-tests to compare all the voices?

One-Way ANOVA

Concept

- Analysis of Variance – compares means across 3+ groups simultaneously
- F-statistic = variance between groups / variance within groups
- Large F – group means differ more than we'd expect by chance

Why ANOVA and not t-tests?

- Running 6 pairwise t-tests inflates the Type I error rate (rejecting the null hypothesis when it is true)
- ANOVA controls this by testing all groups at once

One-Way ANOVA

```
scipy.stats.f_oneway(soprano, alto, tenor, bass)
```

```
— One-Way ANOVA: Mean Melodic Interval across S/A/T/B
— Soprano: M = 1.849, SD = 0.297, n = 320
     Alto: M = 1.703, SD = 0.268, n = 320
    Tenor: M = 1.953, SD = 0.304, n = 320
     Bass: M = 3.122, SD = 0.424, n = 320

     F = 1251.891, p = 0.000000 η² = 0.7464 (η² > 0.06
= medium effect, > 0.14 = large)
```

Conclusion: Reject H_0 — at least one voice differs

Post-hoc Tests: Tukey HSD

- If ANOVA tells us that at least one group differs, which one?
- Tukey Honest Significant Difference tests all pairwise comparisons
- Adjusts for multiple comparisons

Post-hoc Test: Tukey HSD

```
tukey = pairwise_tukeyhsd(interval_vals, voice_labels, alpha=0.05)
```

```
—— Tukey HSD Post-hoc Test ——
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
 group1  group2 meandiff p-adj   lower   upper  reject
-----------------------------------------------------
   Alto    Bass   1.4186    0.0  1.3517  1.4856   True
   Alto Soprano   0.1463    0.0  0.0794  0.2133   True
   Alto   Tenor   0.2503    0.0  0.1834  0.3173   True
   Bass Soprano  -1.2723    0.0 -1.3393 -1.2053   True
   Bass   Tenor  -1.1683    0.0 -1.2353 -1.1013   True
Soprano   Tenor    0.104 0.0004   0.037   0.171   True
-----------------------------------------------------
```

Count data

- What if we want to compare counts and not distributions?
- Example question:
  - Are there more women or men in violin orchestra section vs tuba orchestra section, and is the difference statistically significant?

Type of instrument	Women	Men
Trombone	5	10
Violin	13	9

! We cannot use t-test to compare these groups because these are not distributions!

Chi-squared test (χ^2)

Concept

- Used when both variables are categorical (counts, not means)
- Compares observed frequencies with expected frequencies
- H_0: The two variables are independent (no association)
- $\chi^2 = \Sigma\, [(\text{Observed} - \text{Expected})^2 / \text{Expected}]$
- Larger $\chi^2 \rightarrow$ observed counts deviate more from expectation
- Degrees of freedom = (rows − 1) × (cols − 1)

Musicological example

- Is the presence of accidentals associated with voice type?
- We build a contingency table: voice × accidental counts
- Expected: if voice type doesn't matter, accidentals should be equally proportioned

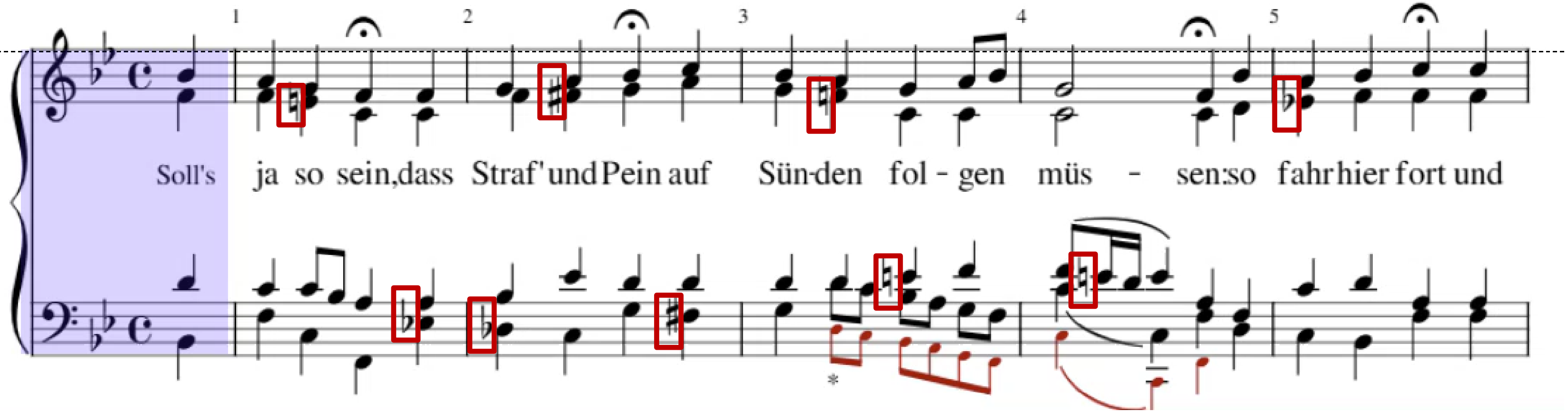
Chorale BWV 48.3

4. Ach Gott und Herr

(Soll's ja so sein, Cantate No. 48. Ich elender Mensch. B.A. 10. S.288, BWV 48.3)

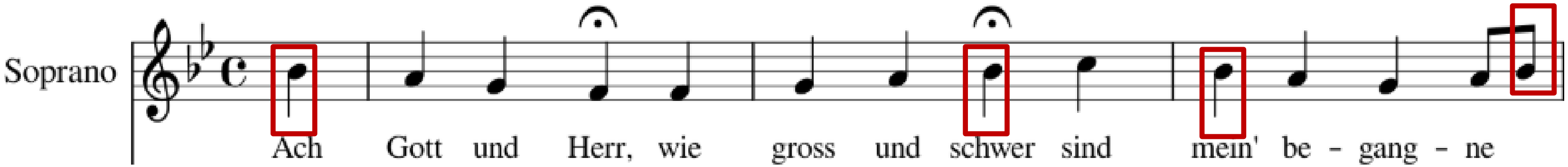
M. Rutilius (1551-1618)

J.S. Bach (Orig. As hymnodus sacer, Leipzig 1625)



Never trust AI written code without checking!

```python
def get_accidental_rate(part):
    """
    Proportion of notes that carry an explicit accidental (chromatic alteration).
    This detects notes that deviate from the prevailing key signature.
    """
    notes = [n for n in part.flatten().notesAndRests if n.isNote]
    if not notes:
        return 0.0
    accidentals = sum(1 for n in notes if n.pitch.accidental is not None)
    return accidentals / len(notes)
```



Second attempt

```
key = score.analyze('key’)


def get_accidentals(part, key):
    """
    Proportion of notes that carry an explicit accidental (chromatic alteration).
    This detects notes that deviate from the prevailing key signature.
    """
    key_pitches = [p.pitchClass for p in key.pitches]
    notes = [n for n in part.flatten().notesAndRests if n.isNote]
    accidental_notes = [
        n for n in notes
        if n.pitch.pitchClass not in key_pitches
    ]
    return len(accidental_notes)
```

Third attempt

```
key = score.analyze('key')
key = score.flat.getElementsByClass('KeySignature')[0]

def get_accidentals(part, key):
    """
    Proportion of notes that carry an explicit accidental (chromatic alteration).
    This detects notes that deviate from the prevailing key signature.
    """
    key_pitches = [p.pitchClass for p in key.pitches]
    notes = [n for n in part.flatten().notesAndRests if n.isNote]
    accidental_notes = [
        n for n in notes
        if n.pitch.pitchClass not in key_pitches
    ]
    return len(accidental_notes)
```

Correlation

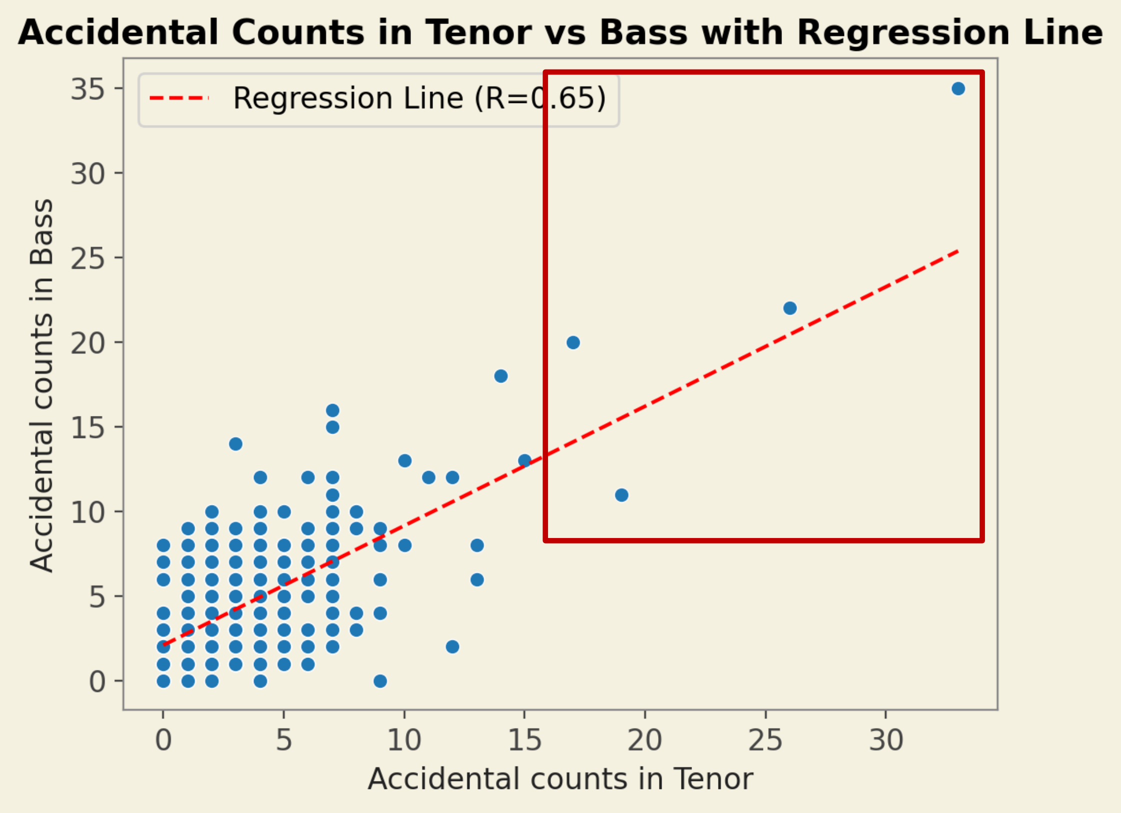
Concept

- Measures strength and direction of linear relationship between two numerical variables
- H_0: No linear relationship between variables ($r = 0$)
- r ranges from -1 to $+1$
- $|r|$ larger → stronger linear relationship
- $r > 0$ → variables increase together
- $r < 0$ → one increases while the other decreases

Musicological example

- In the same song, if an alto has a lot of accidentals, is it likely that a tenor voice would have a lot as well? Are these two variables behaving similarly?

Checking data on a scatterplot

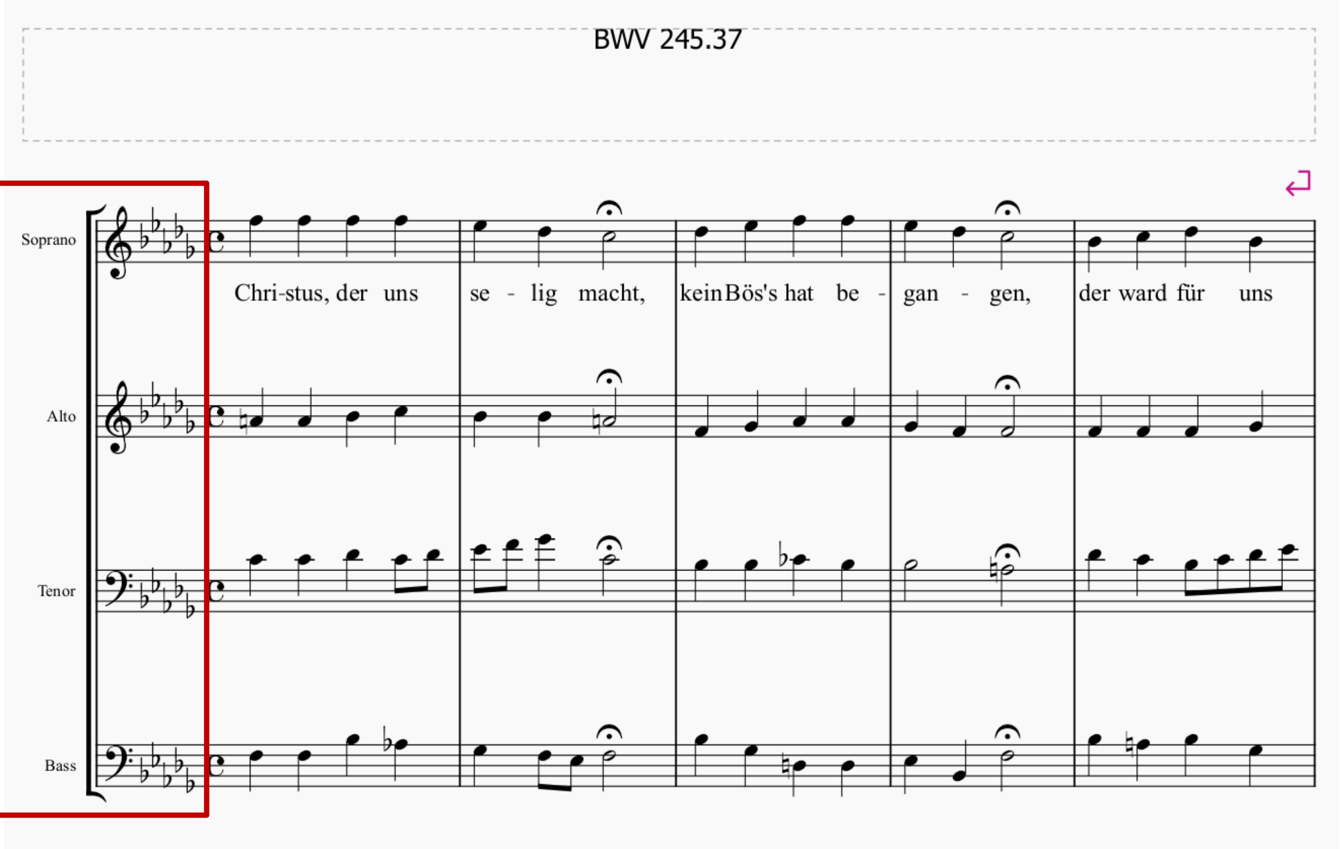


Is it a mistake?

	chorale	chorale_idx	voice	mean_interval	median_interval	sd_interval	max_interval	note_range	accidentals	notes_in_part
314	bach/bwv245.37.mxl	78	Tenor	1.691176	1.0	1.759490	7	10	27	69
439	bach/bwv276.mxl	109	Bass	2.925170	2.0	3.160316	12	22	27	148
649	bach/bwv328.mxl	162	Alto	1.772242	2.0	1.261846	7	14	34	282
650	bach/bwv328.mxl	162	Tenor	2.317164	2.0	2.142360	12	17	26	269
833	bach/bwv371.mxl	208	Alto	1.915179	2.0	1.548888	9	12	35	225
834	bach/bwv371.mxl	208	Tenor	2.137500	2.0	1.693692	9	14	30	241
835	bach/bwv371.mxl	208	Bass	3.050228	2.0	2.876807	12	21	27	220
1125	bach/bwv437.mxl	281	Alto	1.941176	2.0	1.764706	12	15	31	171

Chorale BWV245.37

Chorale BWV245.37



Data cleaning

- Manually checking the suspicious data points
- Removing the outliers when manual checking is infeasible
- Missing values
  - Removing rows with too many
  - Imputation (replacing mean/median)

Preparing a table for chi-squared test

```
df["non_accidentals"]=df["notes_in_part"] - df["accidentals"]
df[df.chorale=="bach/bwv102.7.mxl"][["voice", "accidentals", "non_accidentals"]]
```

	voice	accidentals	non_accidentals
8	Soprano	3	51
9	Alto	8	54
10	Tenor	10	47
11	Bass	9	46

Chi-squared test: results

```
chi2, p_chi2, dof, expected = stats.chi2_contingency(contingency_table_data)


# Cramér's V effect size
n    = contingency_table_data.values.sum()
cramers_v = np.sqrt(chi2 / (n * (min(contingency_table_data.shape) - 1)))
```

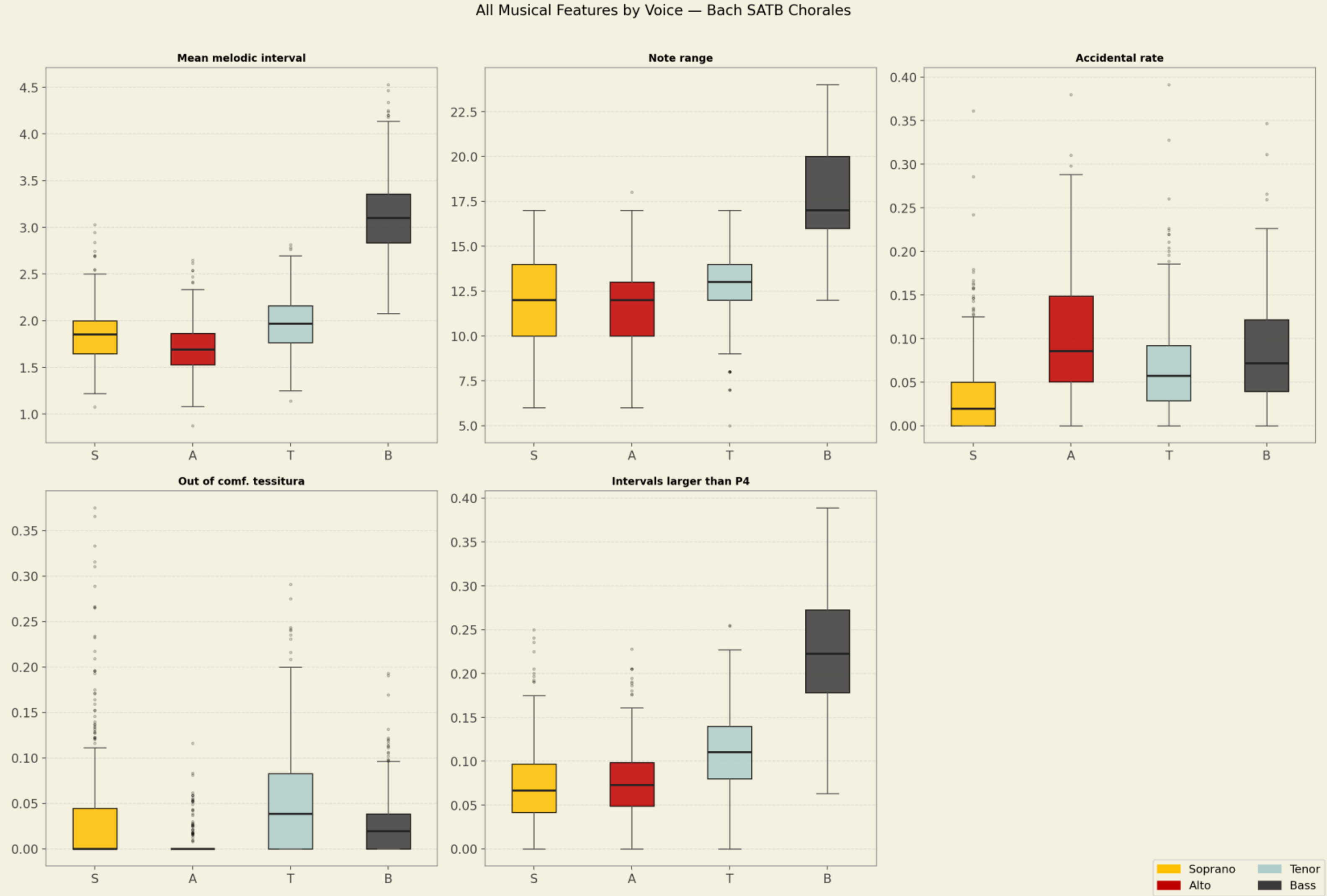
```
── Chi-squared Test ──
χ² = 4.189, df = 3, p = 0.2418

Cramér's V = 0.136 (V > 0.3 = medium, > 0.5 = large)
Conclusion: Fail to reject H₀
Expected frequencies (all should be ≥ 5 for test validity):
        accidentals non_accidentals
Soprano      7.1            46.9
Alto         8.2            53.8
Tenor        7.5            49.5
Bass         7.2            47.8
```

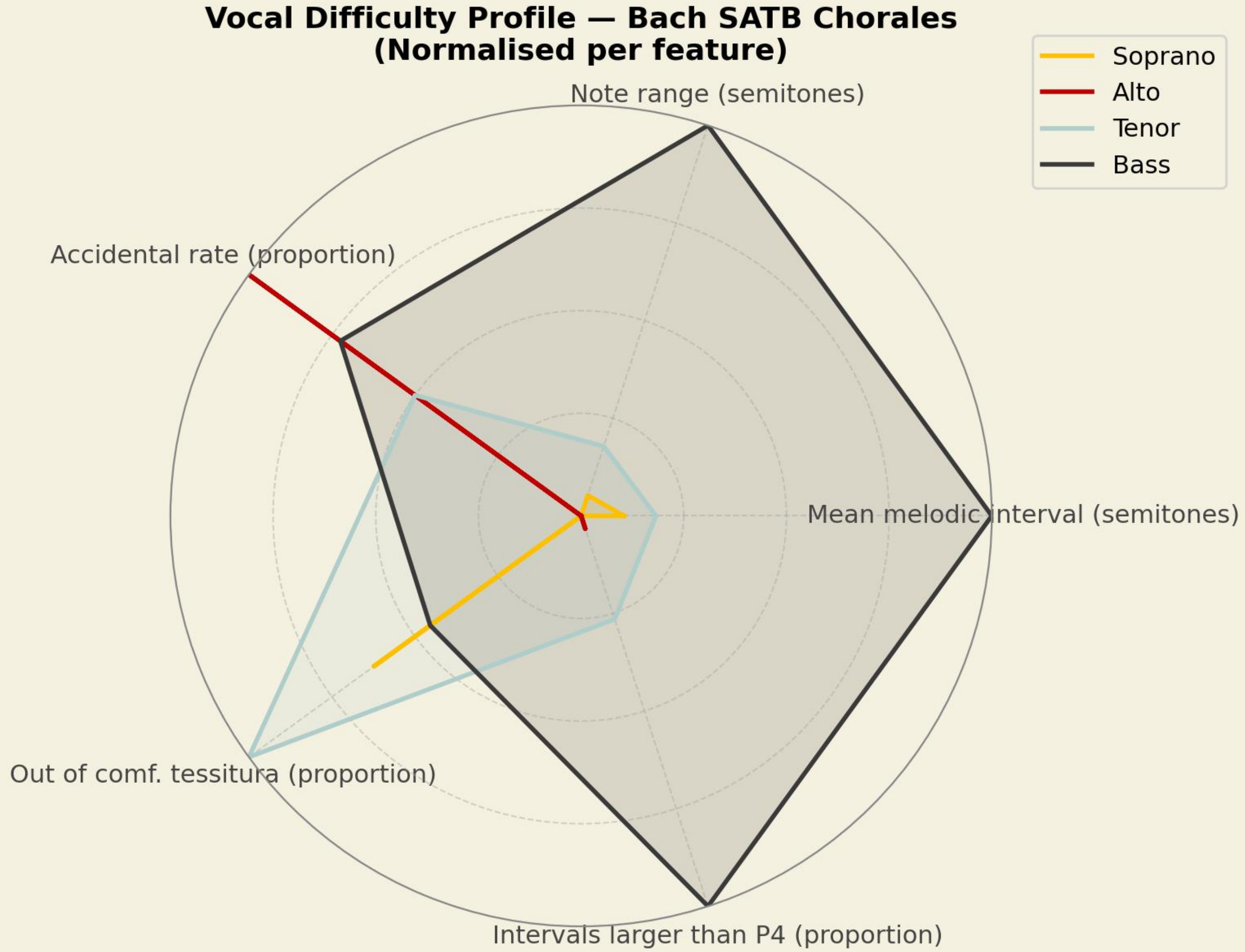
There is a statistical test for every situation

Situation	Test
Expected frequency < 5	Fisher’s exact test
Multiple chi-squared tests across groups necessary	Cochran–Mantel–Haenszel test
Table larger than 2x2	Chi-squared test still works
I have covariates	Mixed effects model
I have paired treatments	McNemar’s test
My categories are ordinal	Cochran-Armitage trend test

Which voice is the hardest?

- Singing out of comfortable tessitura
- Making large melodic jumps
- Dealing with many accidentals
- Covering large vocal range


All Musical Features by Voice — Bach SATB Chorales
Mean melodic interval
Note range
Accidental rate
Out of comf. tessitura
Intervals larger than P4
S
A
T
B
Soprano
Alto
Tenor
Bass


Vocal Difficulty Profile — Bach SATB Chorales
(Normalised per feature)
Soprano
Alto
Tenor
Bass
Note range (semitones)
Accidental rate (proportion)
Mean melodic interval (semitones)
Out of comf. tessitura (proportion)
Intervals larger than P4 (proportion)

Statistical test bingo

Future outlook of jazz students

Dataset Description

The data was collected from **249 students majoring in jazz performence.**

- **Performance Variables:** Preferred musical style versus realistically expected musical style (What style would you like to play? What do you think you will be asked to play at your job?)
- **Financial Variable:** Expected annual income
- **Demographic Variables:** Gender, age, academic status, and ethnicity.

Research Questions

1. **Is there a disparity between students' preferred and expected musical styles?**
2. **Are there differences in expected income based on gender?**
3. **Is age related to preferred style?**
4. **What is the relationship between expected income and age?**